Energizing America

Diversifying the US electricity supply mix

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GQG Research

PDF

Key Takeaways

- Utilities are transitioning from coal to renewables and nuclear energy sources with new reactors coming online for the first time in decades
- The shift to renewables requires three times the generation capacity of coal or natural gas due to their intermittent nature, a transition that comes with its own set of challenges
- While the role of coal is diminishing in power production, natural gas continues to fill the gap when renewable energy sources are not producing at full capacity

Amid the explosive demand for electricity by data centers and clean-technology factories, utilities and regulators are grasping to expand their power supply mix and modernize the country's dated power grid. But utilities are not homogeneous. In our view, to understand what types of utilities are better positioned to navigate the industry obstacles will likely depend on two main characteristics: (1) the asset composition of their fleet; and (2) the political orientation of their primary regulator .

NUCLEAR

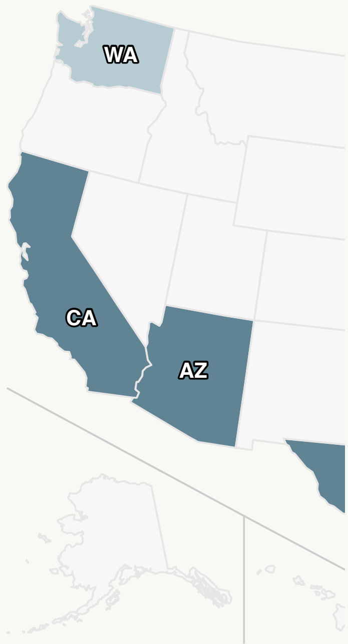
With a limited number of players in the country, the value per nuclear plant is rising. As we see it, companies that own nuclear plants, including Vistra Corp [NYSE: VST], Constellation Energy [NASDAQ: CEG], and

Southern Company [NYSE: SO], are benefiting from the rising demand of low-carbon power supply.

US nuclear power plants generate 775 billion kilowatts-hour (kWh) of electricity, or around 19 percent of the country's total electrical output. It is a scarce asset despite the US being the world's largest producer of nuclear power, accounting for about 30 percent of worldwide generation of nuclear electricity. The country's nuclear capacity is 91.5 gigawatts (GW), which is generated by 93 reactors in 30 of the country's 50 states.

The stocks of independent power producers have reacted to this quite dramatically. Shares of Vistra, a Texas-based power producer with nuclear and natural-gas plants, were up 150 percent for the May year-to-date period on the expectation of additional demand. Forward power prices are also responding.

US Operating Commercial Nuclear Power Rectors



Source: U.S. Nuclear Regulatory

Nuclear power's ability to deliver emissions-free electricity has already garnered demand from hyperscalers like Amazon's AWS, which most recently set the industry's blueprints with Talen Energy [OTC: TLNE] for "behind the meter" deals.

Talen's plant in Pennsylvania will provide almost 1,000 megawatts (MW) to an AWS data center to be built next to Talen's facility. Amazon will pay a 10-year rate of $75-80 per megawatt-hour (MWh) to quickly secure power at a large scale while Talen will be making a 70 percent premium on the deal. For context, this plant would get $45 MWh if it were to sell that power to the market.

FOSSIL FUELS

Demand for natural gas is expected to be more volatile going forward–lower on average, but potentially much higher on peak-demand days when intermittent renewables are at low generation levels. Today's gas systems were not designed and sized to deliver the high gas volumes that will be needed on these peak-demand days in the future. That said, power generation providers will continue to have natural gas as an insurance policy to improve the reliability of the system.

Since about 60 percent of the total electricity generated in the US is from fossil fuels, mostly from natural gas (43 percent) and coal (16 percent), gas utility operators are starting to see demand tailwinds from energy-hungry facilities. Vistra Corp in Texas, Duke Energy [NYSE: DUK] in the Carolinas, and Southern Co in Georgia are natural-gas providers that have benefited from this demand, and all are based in favorable regulatory environments for their businesses, in our opinion.

The US natural gas pipeline network already has about 3 million miles of pipelines that link natural gas production areas and storage facilities with customers. In 2022, this transportation network delivered about 29.2 trillion cubic feet of natural gas to consumers.

There is a broader market recognition of the massive power needs for datacenters and thus the enormous capital expenditure opportunities for gas utilities to provide an alternative energy supply to the grid. US utilities will need to invest around $50 billion in new generation capacity, including new pipelines, as incremental data center power consumption in the US will drive around 3.3 billion cubic feet per day of new natural gas demand by 2030.[1]

Fossil fuels power nearly 60 percent of electricity generation

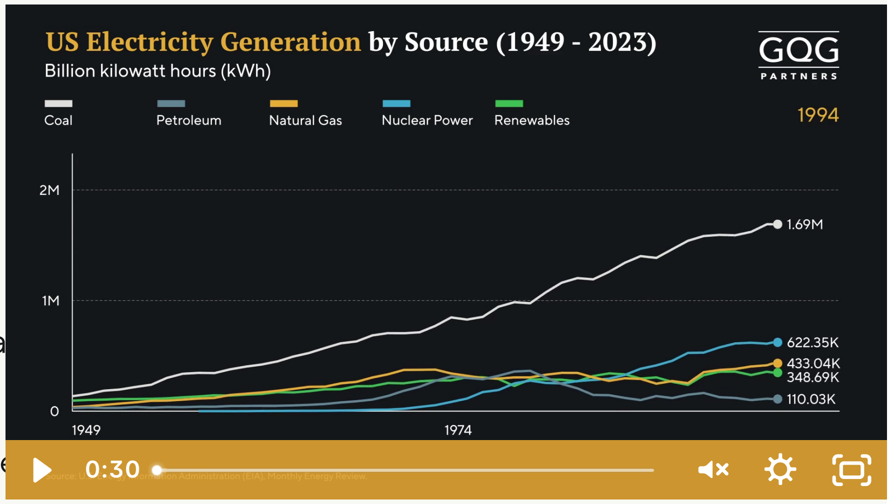
- Gas facility operators saw increased demand from energy-hungry facilities up nearly **40 percent per day** from 2000 to 2022
- Data Center power consumption will drive **3.3 billion cubic feet per day** of new natural gas demand by 2030
- **$50B in new generation capacity** needed to meet the growing demand

RENEWABLES

In 2022, the federal government announced new funding under the Inflation Reduction Act to advance deployment of renewables in the medium term and boost investment in both power plants and equipment manufacturing. Last year, renewables generated more than 21 percent of total electricity in the US. NextEra Energy [NYSE: NEE], one of the largest renewable energy companies in the country, generated around 29 GW of capacity with wind representing 66 percent of the company's generating assets.

In the US, solar power generation has helped to boost the volume of electricity created from renewable sources, with 42 GW of renewable power generating capacity added to the US grid last year. US electricity generation is expected to grow by 3 percent this year, utility-scale solar power will likely contribute nearly 60 percent of this growth, according to the EIA. Wind is expected to comprise 19 percent of the growth and hydropower 13 percent.

However, utilities are going to run into a problem of needing to build many billions of dollars' worth of transmission infrastructure to the remote locations where wind and solar projects are tolerated. "Nobody wants a windmill or solar panels in their backyard."



BATTERY

Battery storage capacity ha
Texas, "but not nearly what
produced by increased rene
regulator. California has the greatest installed battery storage capacity of any state, with 7.3 GW, followed by Texas with 3.2 GW. The remaining 48 states have a combined total of around 3.5 GW of installed battery storage capacity. The EIA expects a significant increase in battery storage capacity over the next three years, reaching 40 GW by the end of 2025.

Vistra Corp. owns the largest battery storage facility in the world, located in California with 750 MW, and developers expect to bring more than 300 utility-scale battery storage projects online by 2025, with half of that capacity in Texas.

US Battery Storage Capacity Expected to Double in 2024